Reference vehicle: Terrier Mk12–Improved Orion

Abstract

This project involved the development of two Python-based simulation tools to analyze multistage rocket flight dynamics and ideal rocket nozzle performance. The flight simulator models vertical ascent using time-dependent thrust, propellant consumption, atmospheric conditions, aerodynamic drag, variable gravity, and stage separation. A separate nozzle simulator applies compressible-flow relationships to evaluate the influence of expansion ratio and ambient pressure on theoretical propulsion performance.

The Terrier Mk12–Improved Orion was selected as the reference vehicle. Numerical verification included analytical comparisons, standard-atmosphere reference values, independent fourth-order Runge–Kutta (RK4) integration, and timestep convergence testing. For the revised configuration, the primary simulation and independent RK4 solution differed by approximately 0.04% in predicted apogee.

Four investigated motor-input configurations produced predicted apogees from approximately 81 to 258 km. This spread greatly exceeds the observed numerical integration differences and demonstrates that uncertainty in vehicle and propulsion inputs dominates the model predictions. Because configuration-specific motor data remain incomplete, these results are illustrative numerical estimates rather than experimentally validated flight predictions.

1. Introduction and Project Scope

Rocket flight performance depends on the interaction of propulsion, aerodynamics, atmospheric conditions, and changing vehicle mass. During ascent these effects continuously influence acceleration, velocity, and altitude. Multistage vehicles introduce additional complexity through propellant depletion, stage separation, and changing geometry. Numerical simulation makes it possible to evaluate these coupled effects and investigate how assumptions influence predicted performance.

The objective was to develop a Python simulation of the vertical flight of a two-stage sounding rocket. Using the Terrier Mk12–Improved Orion as a reference configuration, the flight model computes altitude, velocity, acceleration, dynamic pressure, and vehicle mass through powered ascent and coast to apogee. It includes prescribed thrust curves, changing mass, altitude-dependent gravity, atmospheric properties, Mach-dependent drag, and stage separation.

A separate nozzle simulator evaluates ideal, one-dimensional, isentropic compressible flow. It calculates exhaust conditions, thrust, and specific impulse as functions of expansion ratio and

ambient pressure. The nozzle model is not coupled to the reference flight run, which uses prescribed motor thrust curves.

The flight model does not include launch inclination, wind, Earth rotation, rotational dynamics, or measured motor data. The nozzle model omits flow separation, viscous losses, and other nonideal effects. These simplifications permit transparent numerical verification while limiting direct comparisons with actual vehicle performance.

2. Vehicle and Input Data

2.1 Published descriptions and source differences

NASA's Terrier–Improved Orion vehicle description reports an 18-inch Terrier diameter, a 14-inch Improved Orion diameter, and a vehicle mass of approximately 2,900 lb without payload. It describes Orion thrust of about 19,000 lbf for four seconds, followed by about 3,000 lbf until burnout near 25 seconds. Its payload/altitude description spans roughly 200 km for a 200 lb payload to 80 km for an 800 lb payload [1]. The 2015 handbook states that its performance charts assume sea-level launch at 85° elevation [2].

The current model instead uses approximate Orion thrust levels of 20,000 lbf for six seconds and 4,000 lbf through 24 seconds, alongside a 2,850 lb reference vehicle mass. These figures came from project notes citing a 2023 NASA handbook, but the configuration-specific passage has not been independently confirmed for this report. Differences between descriptions may reflect publication or configuration changes. The values are therefore treated as approximations, not definitive motor measurements.

2.2 Inputs used in the model

Parameter	Value used	Status
Terrier / Orion diameter	0.4572 m / 0.3556 m	NASA vehicle description [1]
Payload	136.08 kg (300 lb)	Selected simulation scenario
Terrier / thrust duration	4.4 s; 257.5 kN, approximately constant	Simplified thrust shape; Secondary source [3]
Terrier propellant / dry mass	578 kg / 322 kg	Secondary source [3]; provisional
Orion thrust	88.96 kN for 0-6 s; 17.79 kN for 6-24 s	Unconfirmed for configuration
Orion propellant mass	378.8 kg	Calculated from impulse using assumed I_{sp}
Orion dry mass	150 kg	Assumption
Specific impulse for mass estimate	230 s	Assumption
Drag coefficient versus Mach	Tabulated in Flight.py	Assumed; not measured
Staging	Instantaneous; no coast	Simplification

Terrier burn time is uncertain: one secondary source reports 4.4 s [3], while another reports approximately 5.2 s
[4]. The original flight model used 5.2 s.

2.3 Mass accounting

The current input set gives a launch mass of approximately 1,564.8 kg, or 1,428.8 kg excluding payload. The original model allocated first-stage dry mass to force the total to equal the 2,850 lb reference. This produced a numerical match without independent physical support and was removed. The revised model is approximately 136 kg heavier than the 2,850 lb reference and 113 kg heavier than the 2,900 lb NASA vehicle description. The stage estimates are not mutually reconciled. If the Terrier motor is 900 kg gross, approximately 415 kg remains within a 2,900 lb vehicle for the Orion and other hardware, whereas the modeled Orion stage alone is about 529 kg. The approximate Terrier impulse of 1.13 MN·s divided by 578 kg of propellant implies an effective specific impulse near 200 s; the Orion propellant estimate instead uses 230 s. A configuration-specific mass breakdown is needed to resolve these inconsistencies.

3. Flight Model

3.1 Equations

Taking upward as positive, the vertical equation of motion is:
$dv/dt = (T + D) / m - g(h)$. Here T is thrust, D is signed aerodynamic drag, m is instantaneous mass, and g(h) is gravitational acceleration.

Drag, gravity, and dynamic pressure are:
$D = -\frac{1}{2} \rho v |v| C_d A$
$g(h) = g_0 [Re/(Re + h)]^2$
$q = \frac{1}{2} \rho v^2$

The model uses $g_0 = 9.80665$ m/s² and Re = 6,356,766 m. Reference area A follows the diameter of the attached stage. For a thrust-curve stage, propellant use during a timestep is proportional to the fraction of total thrust impulse delivered during that step. This assumes constant effective Isp through the burn. Prescribed thrust does not vary with ambient pressure.

3.2 Atmosphere

Below approximately 86 km, the model uses 1976 U.S. Standard Atmosphere layer relationships [5], with geopotential altitude $H = Re\ h/(Re + h)$. At higher altitudes it interpolates tabulated values: linearly for temperature and logarithmically for pressure and density. Density uses the ideal gas law with R = 287.05 J/(kg·K); the speed of sound is estimated as $\sqrt{(\gamma RT)}$ with $\gamma = 1.4$. The continuum speed-of-sound approximation loses physical meaning at very high altitude, where aerodynamic drag is negligible.

3.3 Numerical method and staging

The default timestep is 0.01 s. Forces are evaluated at the beginning of each step, velocity is updated, and altitude is updated using the new velocity. This is a semi-implicit Euler scheme. Thrust over each step is obtained by integrating the piecewise-linear thrust curve, rather than sampling only a single point.

At burnout the spent stage dry mass is removed and the next stage ignites immediately. Following the final burnout, the vehicle coasts with zero thrust until vertical velocity reaches zero. A launch-pad constraint prevents downward motion through the ground when thrust is insufficient for liftoff; recorded acceleration also reflects that constraint.

4. Nozzle Model

4.1 Equations

The nozzle model assumes ideal-gas, isentropic, one-dimensional flow with constant specific-heat ratio γ, a choked throat, and attached flow. Standard compressible-flow relationships [6, 7] give:

$$\dot{m} = P_c A_t \sqrt{\gamma/(R T_c)} \left[2/(\gamma + 1)\right]^{(\gamma + 1)/(2(\gamma - 1))}$$

The exit Mach number is found by bisection on the supersonic branch of the area–Mach relationship:

$$A/A^* = (1/M) \left[(2/(\gamma + 1))(1 + (\gamma - 1)M^2/2)\right]^{(\gamma + 1)/(2(\gamma - 1))}$$

Exit temperature, pressure, and velocity follow from isentropic relations. The performance quantities are:

$$F = \dot{m} V_e + (P_e - P_a) A_e$$

$$I_{sp} = F/(\dot{m} g_0)$$

$$c^* = P_c A_t/\dot{m}$$

$$C_f = F/(P_c A_t)$$

4.2 Inputs and calculated results

The illustrative nozzle inputs are Pc = 3.0 MPa, Tc = 3,400 K, $\gamma = 1.22$, R = 355 J/(kg·K), throat diameter 50 mm, and expansion ratio 8. They do not represent verified Terrier or Orion motor chamber conditions.

Quantity	Result
Throat / exit area	1.963×10^{-3} m^2 / 1.571×10^{-2} m^2
Mass flow rate	3.498 kg/s
Exit Mach number	3.173
Exit pressure	48.10 kPa
Exit velocity	2,652 m/s
Sea-level thrust	8,441 N
Sea-level specific impulse	246.1 s
Characteristic velocity	1,684 m/s
Thrust coefficient	1.433

At sea level the expansion-ratio-8 nozzle is overexpanded: exit pressure is approximately 47.5% of ambient pressure. The sea-level optimum occurs near expansion ratio 4.66, with predicted maximum thrust of 8,653 N. In the modeled 0–10 km altitude sweep, the fixed expansion ratio loses at most 2.46% relative to reoptimization at each altitude. Ideal expansion occurs around 5.8–5.9 km, while the predicted optimum ratio at 10 km is approximately 12.5.

These are ideal attached-flow calculations. The model omits internal flow separation, which can invalidate the predictions under strong overexpansion. It also excludes divergence losses and combustion inefficiency.

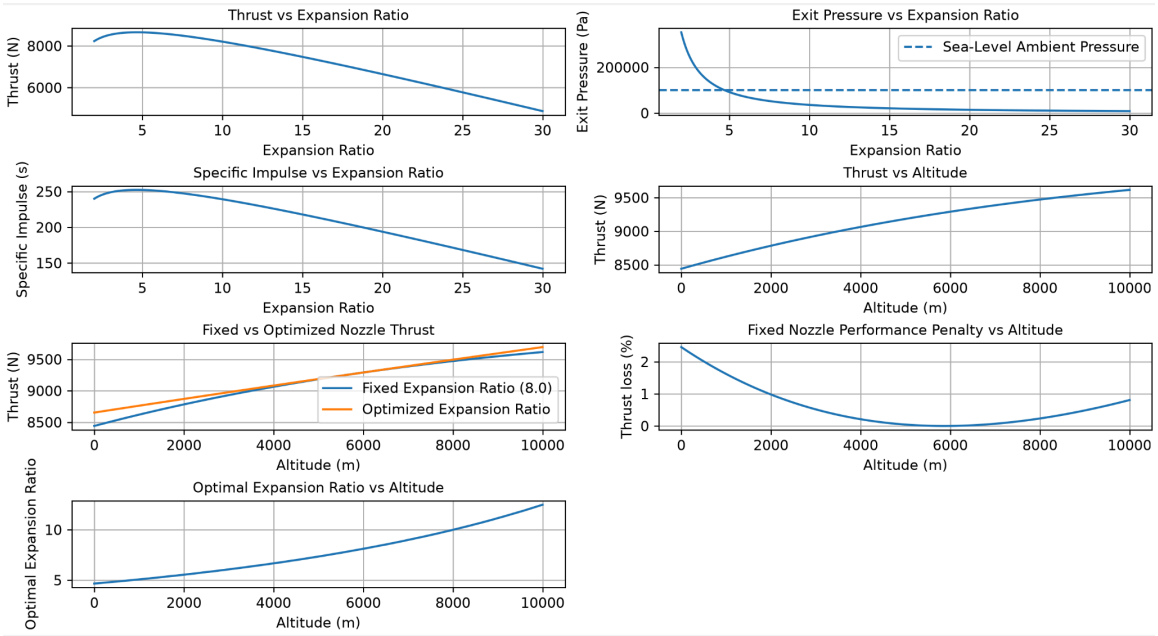


Figure 2 Seven-panel nozzle performance plot. Ideal expansion and altitude trends; flow separation is not modeled.

The nozzle inputs were selected as representative engineering assumptions rather than specifications for a particular rocket motor. Chamber pressure (3.0 MPa) and temperature (3,400 K) were chosen to provide a plausible operating condition for an illustrative chemical rocket nozzle. The specific-heat ratio (1.22) and gas constant (355 J/(kg·K)) approximate hot combustion-gas properties under the ideal-gas assumption. A 50 mm throat diameter establishes a convenient reference geometry, while an expansion ratio of 8 provides a baseline for investigating overexpanded, ideally expanded, and underexpanded operating conditions as ambient pressure changes. These parameters are held constant where appropriate to isolate the effects of nozzle geometry and atmospheric pressure. Their selection is not based on verified Terrier or Orion motor measurements, and the resulting performance values should not be interpreted as predictions for either motor.

5. Verification

5.1 Automated tests

The revised project reports 13 passing tests covering thrust impulse, atmospheric reference values, drag direction, burnout mass, propellant conservation, insufficient-thrust liftoff, analytical rocket-equation comparison, timestep convergence, nozzle area–Mach inversion, ambient-pressure behavior, ideal expansion, and an expansion-condition edge case. The analytical burnout-velocity comparison uses a tolerance of 1 m/s for a simplified short-duration burn.

5.2 Independent numerical checks

Check	Reported result
Pressure vs. standard atmosphere, 0-84.9 km	Within 0.02%
Density vs. standard atmosphere, 0-32 km	Within 0.02%
Revised RK4 vs. flight simulator apogee	258,126 m vs. 258,014 (0.04%)
Original RK4 vs. flight simulator apogee	127,015 m vs. 126,987 m (0.02%)
Apogee for $\Delta t = 0.04, 0.02, 0.01, 0.001$ s	257,677; 257,901; 258,014; 258,115 m

Refining the timestep from 0.01 to 0.001 s changes the revised apogee by approximately 101 m, or 0.04%. This is an observed numerical difference, not a formal upper bound on numerical error. It is small relative to uncertainty in the physical inputs.

5.3 Problems identified and corrected

The independent review identified a circular mass check caused by the first-stage dry-mass allocation, an ambient-pressure formula applied beyond its valid tropospheric range in the nozzle script, an unhandled case when no sampled altitude was ideally expanded, and inconsistent acceleration logging while the rocket remained on the pad. These were addressed in the revised software.

5.4 Verification limitations

The tests establish agreement with selected mathematical relationships; they do not establish that the thrust curves, masses, or drag coefficients match the actual rocket. Some tests remain less stringent than desired: atmosphere tests allow approximately 1% error, and the nozzle edge-case test does not fully exercise the formerly failing code path. Strengthening those tests remains useful for future work.

6. Flight Results

6.1 Comparison of four input configurations

Four input sets were examined. “Old Terrier” denotes the original 66.9 kN, 5.2 s first stage with 154 kg propellant and the earlier mass allocation. “New Terrier” uses the provisional 257.5 kN, 4.4 s first stage. “Model Orion” uses the current approximate thrust curve. “NASA-sheet Orion” uses the alternative published thrust description of 19,000 lbf for four seconds and 3,000 lbf until 25 seconds, with propellant inferred at 230 s specific impulse and an assumed dry mass chosen for a 2,900 lb vehicle total. The NASA-sheet Orion configuration is included solely for sensitivity analysis to evaluate how alternative motor assumptions influence predicted flight performance. Its dry mass was selected to approximate the published vehicle mass rather than derived from independently verified component specifications. Consequently, agreement with published flight performance should not be interpreted as validation of this configuration.

Case	First stage	Second stage	Launch mass	Apogee	Max-Q
A	Old Terrier	Model Orion	1,429 kg	127.0 km	427 kPa
B	New Terrier	Model Orion	1,565 kg	258.0 km	575 kPa
C	New Terrier	NASA-sheet Orion	1,451 kg	191.7 km	616 kPa
D	Old Terrier	NASA-sheet Orion	1,315 kg	80.7 km	327 kPa

Case B is the repository reference run. Its first stage burns out at 4.4 s, approximately 1.71 km altitude and 822 m/s. The second stage burns out at 28.4 s, approximately 39.1 km and 2,029 m/s. Apogee is about 258.0 km at 247 s. Max-Q is approximately 575 kPa at 10.4 s, 8.64 km altitude, and Mach 5.03.

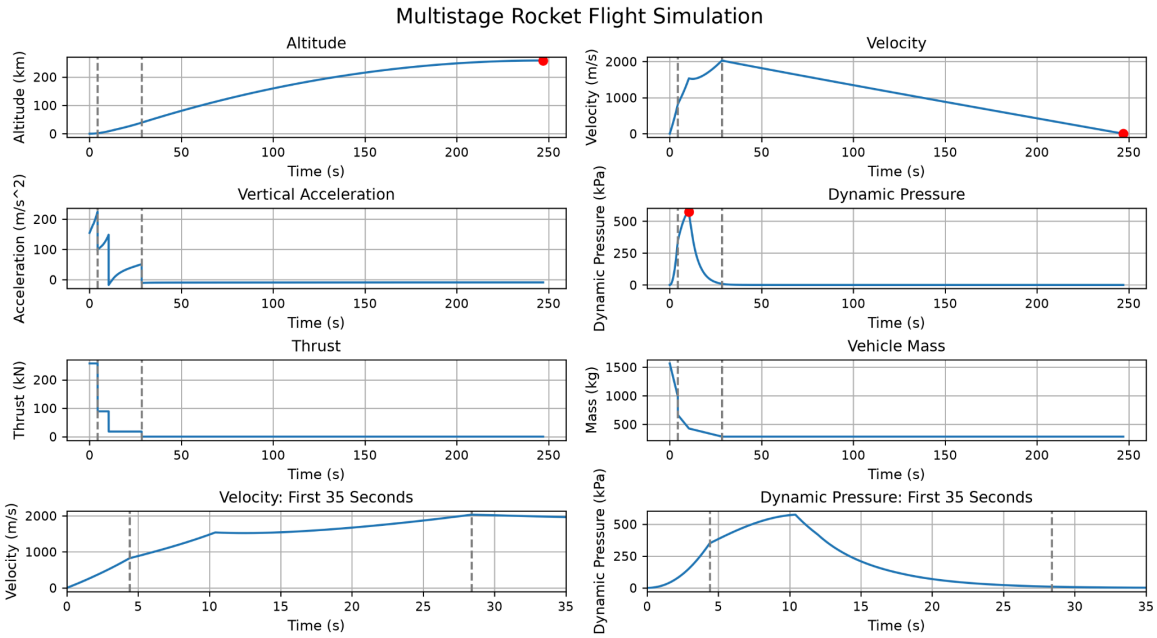


Figure 1: Eight-panel simulated flight plot for case B. Predictions depend on provisional motor and aerodynamic inputs.

6.2 Comparison with published vehicle performance

NASA's published description associates payloads of approximately 200–800 lb with altitudes of roughly 200–80 km [1]. Case B predicts 258 km for a 300 lb payload, exceeding the approximate upper altitude in that reference. This discrepancy suggests that the modeled propulsion, mass, or aerodynamic inputs may not represent the published configuration.

The NASA chart uses an 85° launch elevation while the simulation assumes vertical ascent. A simple geometric estimate suggests that this difference alone is unlikely to explain the full apogee discrepancy, although a rigorous comparison requires multidimensional trajectory modeling. Varying the assumed drag coefficient by ±20% yields apogees of approximately 237–281 km for case B, which also does not resolve the discrepancy.

Cases A and C produce apogees within the broad published range, but such agreement is not evidence of validation: their mass and motor inputs are not independently verified, and aerodynamic assumptions remain provisional. The comparison supports prioritizing configuration-specific motor and aerodynamic data.

6.3 Sensitivity and staging
Across the four cases, predicted apogee spans approximately 81–258 km. A ±20% drag-coefficient variation shifts case B by about ±22 km, whereas timestep changes are below 0.1%. The dominant uncertainty is therefore the selected motor and mass inputs, followed by aerodynamics and then numerical discretization.

In the original configuration, a four- to six-second coast between stages reduced Max-Q from 427 kPa to approximately 365–383 kPa while changing apogee by less than 1 km. Across the four cases, Max-Q ranges from 327 to 616 kPa. Because staging and drag remain simplified, these load predictions should be treated cautiously.

7. Limitations

Motor data: Terrier thrust, duration, and mass values are secondary-source estimates, and the Orion thrust approximations are rounded; neither stage has a configuration-matched measured thrust curve in the model.

Mass budget: Component masses do not reconcile with the NASA vehicle reference mass.

One-dimensional motion: Launch inclination, winds, and rotational dynamics are omitted.

Aerodynamics: The assumed drag coefficient table does not distinguish powered from unpowered base drag.

Staging: Separation and next-stage ignition are instantaneous.

Propulsion: Thrust follows fixed curves rather than altitude-dependent chamber and nozzle conditions.

Nozzle: The ideal, attached-flow calculation is separate from the flight model.

8. Future Work

The primary next step is to obtain configuration-specific propulsion and vehicle data, including thrust-time histories, loaded and inert motor masses, propellant mass, stage timing, and documented flight performance. Additional NASA documentation and flight reports may help reconcile the present mass discrepancies and establish more defensible model inputs.

Future numerical improvements include a staging coast interval, separate powered and unpowered drag coefficients, and a higher-order integration method such as RK4. These changes would improve the representation of stage transitions and aerodynamic loading, but would not replace the need for reliable physical inputs.

A planned experimental extension involves CAD modeling of several nozzle expansion ratios, manufacturing selected geometries, and conducting controlled cold-flow tests. The objective is to compare measured flow and pressure behavior with theoretical nozzle trends. Cold-flow tests would not reproduce hot rocket-motor thermochemistry. This experimental phase has not yet begun.

9. Engineering Findings and Lessons Learned

The project demonstrated that numerical consistency does not guarantee physical accuracy. The initial model ran successfully and passed its original tests, yet a mass allocation had been chosen to force agreement with a published vehicle mass. Removing that allocation exposed the lack of independent support for the stage mass distribution.

Revising the motor inputs changed predicted apogee from approximately 127 km to 258 km without changing the governing flight equations. This showed that uncertainty in thrust and mass can dominate the final result. Analytical comparisons, atmosphere checks, timestep convergence, and independent integration increased confidence in the numerical implementation, but could not establish that the assumed vehicle matches the real Terrier Mk12–Improved Orion.

The principal engineering lesson is that simulation outputs must be presented alongside their assumptions and uncertainty. Numerical precision should not be mistaken for validated physical performance.

10. Conclusion

This project produced two Python-based engineering tools: a vertical multistage flight simulator and an ideal nozzle performance model. The flight simulator represents changing thrust, mass, atmospheric conditions, drag, gravity, and staging through apogee. The nozzle simulator evaluates compressible-flow behavior and the effect of expansion ratio and ambient pressure on theoretical propulsion performance.

Automated testing, analytical comparisons, atmospheric reference checks, independent RK4 integration, and timestep refinement support the numerical consistency of the implementation. For the revised configuration, the predicted apogee was approximately 258 km, differing from an independent RK4 result by about 0.04%. This observed numerical agreement does not establish physical validation.

Four investigated configurations predicted apogees from approximately 81 to 258 km, illustrating the influence of uncertain motor and mass assumptions. The revised case also exceeded the broad published performance reference, underscoring the need for verified propulsion and aerodynamic inputs. The completed software is therefore best understood as a functioning engineering analysis framework, with future work directed toward better vehicle data and planned physical nozzle testing.

References

[1] NASA Wallops Flight Facility. Terrier-Improved Orion (41.XXX), General. https://sites.wff.nasa.gov/code810/vehicles/Terrier_Imrprove%20Orion.pdf

[2] NASA. NASA Sounding Rockets User Handbook, 810-HB-SRP, July 2015, §3.2.1. https://www.nasa.gov/wp-content/uploads/2023/09/sounding-rocket-program-handbook.pdf

[3] Encyclopedia Astronautica. Terrier Mk12 Mod 1. http://www.astronautix.com/t/terriermk12mod1.html (secondary source).

[4] Wikipedia. Terrier Malemute. https://en.wikipedia.org/wiki/Terrier_Malemute (secondary source).

[5] NOAA, NASA, and USAF. U.S. Standard Atmosphere, 1976.

[6] Sutton, G. P., and Biblarz, O. Rocket Propulsion Elements. Wiley.

[7] Anderson, J. D. Modern Compressible Flow: With Historical Perspective. McGraw-Hill.